Question 1: For the reaction A→ProductsA→Products, the rate constant is 5.0×10−2 Ms−15.0×10−2Ms−1. The reaction follows the integrated rate law:

- a. ln[A]t=−kt+ln[A]0ln[A]t=−kt+ln[A]0
- b. [A]t=−kt+[A]0[A]t=−kt+[A]0
- c. ln([A]t[A]0)=ktln([A]0[A]t)=kt
- d. 1[A]t2=kt+1[A]02[A]t21=kt+[A]021
- e. 1[A]t=kt+1[A]0[A]t1=kt+[A]01

Question 2: Which of the following best describes all the intermolecular forces exhibited by a pure sample of HOCH2CH2OHHOCH2CH2OH?

- a. Dipole-dipole and hydrogen bonding
- b. Dispersion and dipole-dipole
- c. Dispersion only
- d. Dispersion, dipole-dipole, and hydrogen bonding
- e. Dispersion and hydrogen bonding

Question 3: Consider the following pairs of liquids. Which pairs are **miscible**?

1. Benzene (C6H6C6H6) and hexane (C6H12C6H12)
2. Water and methanol (CH3OHCH3OH)
3. Water and hexane

- a. 2 only
- b. 1 only
- c. 1, 2 only
- d. 1, 2, 3
- e. 2, 3 only

Question 4: Which one of the following liquids would be expected to have the highest viscosity at room temperature?

- a. C6H6C6H6
- b. CH2Cl2CH2Cl2
- c. HOCH2CH2CH2OHHOCH2CH2CH2OH
- d. HOCH2CH2CH2CH3HOCH2CH2CH2CH3

Question 5: A substance at a temperature greater than its critical temperature is called:

- a. Solid
- b. Liquid
- c. Vapor
- d. Supercritical liquid

Question 6: Rank the following compounds from weakest intermolecular forces to strongest intermolecular forces:

1. H_2OH_2O
2. N_2N_2
3. H_2SH_2S

- **Answer:** $N_2<H_2S<H_2ON_2<H_2S<H_2O$

Question 7: Which of the following phase changes is(are) endothermic?

1. Melting
2. Vaporization
3. Sublimation
4. Condensation
5. Deposition
6. Freezing

- a. 4 and 6 only
- b. 1 and 2 only
- c. 4, 5, and 6
- d. 1, 2, and 3

Question 8: The following graph shows the kinetics curves for the reaction of oxygen with hydrogen to form water:

$O_2(g)+2H_2(g)\rightarrow 2H_2O(g).O_2(g)+2H_2(g)\rightarrow 2H_2O(g)$.

Which curve represents hydrogen ($H_2(g)H_2(g)$)?

- a. The black curve
- b. The gray curve
- c. Either the gray or the black curve
- d. Any of these curves could be hydrogen

- e. The dashed curve

Question 9: A 250 mL solution containing 10.0 g of a polymer in toluene had an osmotic pressure of 0.055 atm at 27°C. What is the molar mass of the polymer?

- a. 18000 g/mol
- b. 38000 g/mol
- c. 15000 g/mol
- d. 26000 g/mol
- e. 32000 g/mol

Question 10: What types of intermolecular forces exist between NH3NH3 and H2OH2O?

- a. Dispersion forces, hydrogen bonds, and ion-dipole forces
- b. Dispersion forces
- c. Dispersion forces and ion-dipole forces
- d. Dispersion forces and hydrogen bonds
- e. Dispersion forces, dipole-dipole forces, and hydrogen bonds

Question 11: Which probably has the lowest boiling point at 1.00 atm?

- a. H2TeH2Te
- b. H2SH2S
- c. H2SeH2Se
- d. H2OH2O

Question 12: Given the following data, determine the rate law for the reaction X+Y→2Z*X*+*Y*→2*Z*:

Exp	**[X] (M)**	**[Y] (M)**	**Rate (M/s)**
1	0.0300	0.0100	3.4×10−43.4×10−4
2	0.0150	0.0100	8.5×10−58.5×10−5
3	0.0150	0.0400	3.4×10−43.4×10−4

- a. Rate = k[X]2[Y]*k*[*X*]2[*Y*]
- b. Rate = k[X][Y]*k*[*X*][*Y*]
- c. Rate = k[X]2[Y]2*k*[*X*]2[*Y*]2
- d. Rate = k[X][Y]2*k*[*X*][*Y*]2

Question 13: Which of the following liquids will have the lowest freezing point?

- a. Aqueous CoI2CoI2 (0.030 m)
- b. Pure H2OH2O
- c. Aqueous CaH12O6CaH12O6 (0.60 m)
- d. Aqueous FeI3FeI3 (0.24 m)
- e. Aqueous KFKF (0.50 m)

Question 14: Which of the following choices has the compounds correctly arranged in order of increasing solubility in water? (least soluble to most soluble)

- a. CCl4<CHCl3<CH3OHCCl4<CHCl3<CH3OH
- b. CHCl3<CCl4<NaNO3CHCl3<CCl4<NaNO3
- c. CH3OH<CH4<LiFCH3OH<CH4<LiF
- d. CH4<NaNO3<CHCl3CH4<NaNO3<CHCl3

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14
A	D	C	C	D	$N_2 < H_2S < H_2O$	D	E	B	E	B	D	D	A